

Neeraj Roy

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EXPERIENCE

Software Engineer

July 2019 – Present

Tata Consultancy Services Limited

Gurugram, Haryana

- Leading development of a FastAPI-based backend system for internal inventory tracking, integrating authentication, background tasks, and REST API best practices.
- Designed and implemented PostgreSQL schema with async ORM(SQLALCHEMY 2.0 + Alembic) to support scalable and maintainable storage architecture.
- Led Java and .NET migration and remediation projects, transitioning monolithic frameworks to a microservices architecture, improving deployment, maintainability, and CI/CD readiness.
- Implemented Redis-based caching and Celery/RQ background workers to handle scheduled reminders and email processing at high throughput.
- Assisted in proposing FoxPro and CI/CD legacy applications into a newer ecosystem, implementing fault tolerance and logging to enhance system reliability.

Software Engineering Intern

Aug 2018 – March 2019

RobonomicsAI

GLA University, Mathura

- Built a Chatbot MVP integrated with Telegram Bot and Slack to evaluate chatbot adoption for business solutions.
- Designed and deployed web scrapers to gather and analyze data from various sources using BeautifulSoup and Scrapy.
- Implemented NLP techniques to enhance chatbot capabilities and improve user interactions.

PROJECTS

Qeats | *Jackson, Spring Boot, Spring Data, Rest API, Mockito, JUnit*

- Built different parts of the Qeats backend which is a Spring Boot application.
- Several REST API endpoints are implemented to query restaurant information and place food orders.
- Improved the app performance under large load scenarios as well as included an advanced search feature in the app.

FaceInteract | *Python, Numpy, openCV, PyAutoGUI, Dlib, Imutils, Google SpeechtoText*

- Developed a full desktop application using python to facilitate speech synthesis as keyboard alternative and facial landmark detection for mouse simulation.
- Implement Facial Landmark Detection using Dlib's Prebuild Model and algorithm Histograms of Oriented Gradients.

TripScanner | *Flask, MongoDB, Scrapy, Redis, Git, AWS*

- Developed a Flask WebApp that upon user's input on itinerary details and number of days and budget, scrapes famous offerings based on Google SEO Ranking and presents them in a single page.
- Application used pre-configured template for quick scanning and cached it in db.

EDUCATION

GLA University

Mathura, Uttar Pradesh

B. Tech Computer Science Engineering

Aug. 2015 – July 2019

SKILLS

Languages: Python, C++, Java

Backend Frameworks: FastAPI, Flask, .NET Core, Spring Boot

Databases: PostgreSQL, MongoDB, Redis

Cloud & Devops: AWS(EC2, Lambda, S3) Docker, CI/CD(Github Actions)

AI & Data Engineering: NumPy, Pandas, OpenCV, Whisper, Hugging Face Transformers

GenAI & Agent framework: LangChain, Agno Framework, CrewAI, OpenAI Assistants API, Google GenAI (Gemini), LlamaIndex

Architecture Patterns: Microservices, Clean Architecture, Domain-driven Design, REST APIs

Authentication & Security: OAuth2.0, JWT, RBAC, API rate limiting **Tools & Libraries:** Git, Pydantic, SQLAlchemy, Requests, Scrapy, Pytest, Celery/RQ

Testing: Unit Testing, Integration testing, Mocking, TDD/BDD